ML Lab 5-Aim: To implement the Random Forest algorithm and evaluate the model performance using appropriate metrics.

```
import pandas as pd
from sklearn.model_seletion import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix
```

```
from google.colab import drive
drive.mount('/content/drive')
```

```
Mounted at /content/drive
```

```
path = '/content/drive/MyDrive/ML Dataset/banking - banking.csv'
df=pd.read_csv(path)
df.head()
```

	age	job	marital	education	default	housing	loan	contact	month	day_of_week	...	campaign	pdays	previous
0	44	blue-collar	married	basic.4y	unknown	yes	no	cellular	aug	thu	...	1	999	0
1	53	technician	married	unknown	no	no	no	cellular	nov	fri	...	1	999	0
2	28	management	single	university.degree	no	yes	no	cellular	jun	thu	...	3	6	2
3	39	services	married	high.school	no	no	no	cellular	apr	fri	...	2	999	0
4	55	retired	married	basic.4y	no	yes	no	cellular	aug	fri	...	1	3	1

5 rows × 21 columns

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 41188 entries, 0 to 41187
Data columns (total 21 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   age             41188 non-null  int64
 1   job             41188 non-null  object
 2   marital         41188 non-null  object
 3   education       41188 non-null  object
 4   default         41188 non-null  object
 5   housing         41188 non-null  object
 6   loan            41188 non-null  object
 7   contact         41188 non-null  object
 8   month           41188 non-null  object
 9   day_of_week     41188 non-null  object
 10  duration        41188 non-null  int64
 11  campaign        41188 non-null  int64
 12  pdays           41188 non-null  int64
 13  previous        41188 non-null  int64
 14  poutcome        41188 non-null  object
 15  emp_var_rate    41188 non-null  float64
 16  cons_price_idx  41188 non-null  float64
 17  cons_conf_idx   41188 non-null  float64
 18  euribor3m       41188 non-null  float64
 19  nr_employed     41188 non-null  float64
 20  y               41188 non-null  int64
dtypes: float64(5), int64(6), object(10)
memory usage: 6.6+ MB
```

```
df.describe()
```

	age	duration	campaign	pdays	previous	emp_var_rate	cons_price_idx	cons_conf_idx	euribor3
count	41188.00000	41188.000000	41188.000000	41188.000000	41188.000000	41188.000000	41188.000000	41188.000000	41188.00000
mean	40.02406	258.285010	2.567593	962.475454	0.172963	0.081886	93.575664	-40.502600	3.62129
std	10.42125	259.279249	2.770014	186.910907	0.494901	1.570960	0.578840	4.628198	1.73444
min	17.00000	0.000000	1.000000	0.000000	0.000000	-3.400000	92.201000	-50.800000	0.63400
25%	32.00000	102.000000	1.000000	999.000000	0.000000	-1.800000	93.075000	-42.700000	1.34400
50%	38.00000	180.000000	2.000000	999.000000	0.000000	1.100000	93.749000	-41.800000	4.85700
75%	47.00000	319.000000	3.000000	999.000000	0.000000	1.400000	93.994000	-36.400000	4.96100
max	98.00000	4918.000000	56.000000	999.000000	7.000000	1.400000	94.767000	-26.900000	5.04500

```
df.shape
```

```
(41188, 21)
```

```
df.columns
```

```
Index(['age', 'job', 'marital', 'education', 'default', 'housing', 'loan',
       'contact', 'month', 'day_of_week', 'duration', 'campaign', 'pdays',
       'previous', 'poutcome', 'emp_var_rate', 'cons_price_idx',
       'cons_conf_idx', 'euribor3m', 'nr_employed', 'y'],
      dtype='object')
```

```
X = df.drop(columns=['y'])
y = df['y']
```

```
X.head()
```

	age	job	marital	education	default	housing	loan	contact	month	day_of_week	duration	campaign	pdays	prev
0	44	blue-collar	married	basic.4y	unknown	yes	no	cellular	aug	thu	210	1	999	
1	53	technician	married	unknown	no	no	no	cellular	nov	fri	138	1	999	
2	28	management	single	university.degree	no	yes	no	cellular	jun	thu	339	3	6	
3	39	services	married	high.school	no	no	no	cellular	apr	fri	185	2	999	
4	55	retired	married	basic.4y	no	yes	no	cellular	aug	fri	137	1	3	

```
print(X)
```

```
       age          job  marital            education  default housing loan  \
0       44  blue-collar  married             basic.4y  unknown     yes   no
1       53   technician  married              unknown       no      no   no
2       28   management   single    university.degree       no     yes   no
3       39     services  married          high.school       no      no   no
4       55      retired  married             basic.4y       no     yes   no
...    ...          ...      ...                  ...      ...     ...  ...
41183   59      retired  married          high.school  unknown      no  yes
41184   31    housemaid  married             basic.4y  unknown      no   no
41185   42       admin.   single    university.degree  unknown     yes  yes
41186   48   technician  married  professional.course       no      no  yes
41187   25      student   single          high.school       no      no   no

         contact month day_of_week  duration  campaign  pdays  previous  \
0       cellular   aug         thu       210         1    999         0
1       cellular   nov         fri       138         1    999         0
2       cellular   jun         thu       339         3      6         2
3       cellular   apr         fri       185         2    999         0
4       cellular   aug         fri       137         1      3         1
...          ...   ...         ...       ...       ...    ...       ...
41183  telephone   jun         thu       222         1    999         0
41184  telephone   may         thu       196         2    999         0
41185  telephone   may         wed        62         3    999         0
41186  telephone   oct         tue       200         2    999         0
41187  telephone   may         fri       112         4    999         0

          poutcome  emp_var_rate  cons_price_idx  cons_conf_idx  euribor3m  \
0      nonexistent           1.4          93.444          -36.1      4.963
1      nonexistent          -0.1          93.200          -42.0      4.021
```

```
2          success          -1.7          94.055          -39.8      0.729
3      nonexistent          -1.8          93.075          -47.1      1.405
4          success          -2.9          92.201          -31.4      0.869
...            ...           ...             ...            ...        ...
41183  nonexistent           1.4          94.465          -41.8      4.866
41184  nonexistent           1.1          93.994          -36.4      4.860
41185  nonexistent           1.1          93.994          -36.4      4.857
41186  nonexistent          -3.4          92.431          -26.9      0.742
41187  nonexistent           1.1          93.994          -36.4      4.859

       nr_employed
0           5228.1
1           5195.8
2           4991.6
3           5099.1
4           5076.2
...            ...
41183       5228.1
41184       5191.0
41185       5191.0
41186       5017.5
41187       5191.0

[41188 rows x 20 columns]
```

```
#X = pd.get_dummies(X)
```

```
X = pd.get_dummies(X, dtype = int)
```

```
X.head()
```

	age	duration	campaign	pdays	previous	emp_var_rate	cons_price_idx	cons_conf_idx	euribor3m	nr_employed	...	month_oct
0	44	210	1	999	0	1.4	93.444	-36.1	4.963	5228.1	...	0
1	53	138	1	999	0	-0.1	93.200	-42.0	4.021	5195.8	...	0
2	28	339	3	6	2	-1.7	94.055	-39.8	0.729	4991.6	...	0
3	39	185	2	999	0	-1.8	93.075	-47.1	1.405	5099.1	...	0
4	55	137	1	3	1	-2.9	92.201	-31.4	0.869	5076.2	...	0

5 rows × 63 columns

```
X.shape
```

```
(41188, 63)
```

```
X.columns
```

```
Index(['age', 'duration', 'campaign', 'pdays', 'previous', 'emp_var_rate',
       'cons_price_idx', 'cons_conf_idx', 'euribor3m', 'nr_employed',
       'job_admin.', 'job_blue-collar', 'job_entrepreneur', 'job_housemaid',
       'job_management', 'job_retired', 'job_self-employed', 'job_services',
       'job_student', 'job_technician', 'job_unemployed', 'job_unknown',
       'marital_divorced', 'marital_married', 'marital_single',
       'marital_unknown', 'education_basic.4y', 'education_basic.6y',
       'education_basic.9y', 'education_high.school', 'education_illiterate',
       'education_professional.course', 'education_university.degree',
       'education_unknown', 'default_no', 'default_unknown', 'default_yes',
       'housing_no', 'housing_unknown', 'housing_yes', 'loan_no',
       'loan_unknown', 'loan_yes', 'contact_cellular', 'contact_telephone',
       'month_apr', 'month_aug', 'month_dec', 'month_jul', 'month_jun',
       'month_mar', 'month_may', 'month_nov', 'month_oct', 'month_sep',
       'day_of_week_fri', 'day_of_week_mon', 'day_of_week_thu',
       'day_of_week_tue', 'day_of_week_wed', 'poutcome_failure',
       'poutcome_nonexistent', 'poutcome_success'],
      dtype='object')
```

```
from sklearn.model_selection import train_test_split
X_train, X_test,y_train,y_test = train_test_split(X, y, train_size=0.7, test_size= 0.3,random_state=42)

X_train.shape, X_test.shape, y_train.shape, y_test.shape
```

```
((28831, 63), (12357, 63), (28831,), (12357,))
```

```
from sklearn.ensemble import RandomForestClassifier
rf_classifier = RandomForestClassifier(n_estimators=100, n_jobs=-1, max_depth=5,oob_score=True,random_state=42)
rf_classifier.fit(X_train,y_train)
```

```
RandomForestClassifier
RandomForestClassifier(max_depth=5, n_jobs=-1, oob_score=True, random_state=42)
```

```
y_pred = rf_classifier.predict(X_test)
```

```
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix
```

```
print('Confusion Matrix:')
print(confusion_matrix(y_test, y_pred))
```

```
Confusion Matrix:
[[10848    83]
 [ 1158   268]]
```

```
print('Accuracy Score:')
print(accuracy_score(y_test, y_pred))
```

```
Accuracy Score:
0.8995710933074371
```

```
rf_classifier.oob_score_
```

```
0.9024660955221809
```

```
print('Classification Report:')
print(classification_report(y_test, y_pred))
```

```
Classification Report:
              precision    recall  f1-score   support

           0       0.90      0.99      0.95     10931
           1       0.76      0.19      0.30      1426

    accuracy                           0.90     12357
   macro avg       0.83      0.59      0.62     12357
weighted avg       0.89      0.90      0.87     12357
```

Hypertunning Parameter

```
rf = RandomForestClassifier(random_state=42, n_jobs=-1)
params= {
    'max_depth': [2,3,5,10,20],
    'min_samples_leaf': [5,10,20,50,100,200],
    'n_estimators': [10,25,30,50,100,200]
}
```

```
from sklearn.model_selection import GridSearchCV
grid_search = GridSearchCV(estimator=rf, param_grid=params, cv=4, n_jobs=-1, verbose=1, scoring="accuracy")
grid_search = grid_search.fit(X_train, y_train)
```

```
Fitting 4 folds for each of 180 candidates, totalling 720 fits
```

```
grid_search.best_score_
```

```
np.float64(0.9139467076496568)
```

```
print(grid_search.best_params_)
```

```
{'max_depth': 20, 'min_samples_leaf': 5, 'n_estimators': 200}
```